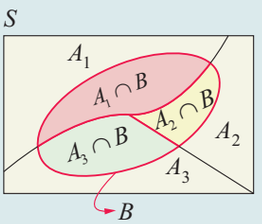
Example 12.28

The chances of X, Y and Z becoming managers of a certain company are 4 : 2 : 3. The probabilities that bonus scheme will be introduced if X, Y and Z become managers are 0.3, 0.5 and 0.4 respectively. If the bonus scheme has been introduced, what is the probability that Z was appointed as the manager?

Solution

Let A_1, A_2 and A_3 be the events of X, Y and Z becoming managers of the company respectively. Let B be the event that the bonus scheme will be introduced.



We have to find the conditional probability $P(A_3 / B)$.

Since A_1, A_2 and A_3 are mutually exclusive and exhaustive events, applying Bayes' theorem

We have
$$P(A_3 / B) = \frac{P(A_3)\,P(B / A_3)}{P(A_1)\,P(B / A_1) + P(A_2)\,P(B / A_2) + P(A_3)\,P(B / A_3)}$$

$$P(A_1) = \frac{4}{9}, \quad P(B / A_1) = 0.3$$

$$P(A_2) = \frac{2}{9}, \quad P(B / A_2) = 0.5$$

$$P(A_3) = \frac{3}{9}, \quad P(B / A_3) = 0.4$$

$$P(A_3 / B) = \frac{P(A_3)\,P(B / A_3)}{P(A_1)\,P(B / A_1) + P(A_2)\,P(B / A_2) + P(A_3)\,P(B / A_3)}$$

$$P(A_3 / B) = \frac{\left(\frac{3}{9}\right)(0.4)}{\left(\frac{4}{9}\right)(0.3) + \left(\frac{2}{9}\right)(0.5) + \left(\frac{3}{9}\right)(0.4)}$$

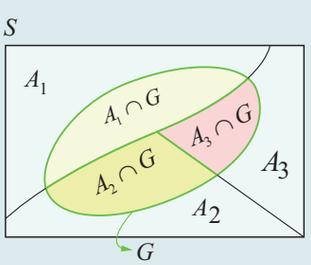
$$= \frac{12}{34} = \frac{6}{17}.$$

Example 12.29

A consulting firm rents car from three agencies such that 50% from agency L, 30% from agency M and 20% from agency N. If 90% of the cars from L, 70% of cars from M and 60% of the cars from N are in good conditions (i) what is the probability that the firm will get a car in good condition? (ii) if a car is in good condition, what is probability that it has come from agency N?

Solution

Let $A_1, A_2,$ and A_3 be the events that the cars are rented from the agencies L, M and N respectively.



Let G be the event of getting a car in good condition.

We have to find

(i) the total probability of event G that is, $P(G)$

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